SAKU Election System

Secure electronic election management for the Students Association of KCA University

Not an online form for collecting votes. A command interface for administering a two-stage student election — with authority, audit, and a controlled ballot.

PROTOTYPE · VALIDATION BUILD · 2026

THE PROBLEM

Student elections need more than a survey.

What breaks with ordinary online forms

- Anyone can be shown every candidate list
- Login success is treated as full power
- No station-level election officer record
- Results leak as public percentages
- No auditable pause, opening, or declaration

What SAKU actually requires

- A large-electorate delegate election, then a closed Cabinet election
- Official campus / school / department structure
- Commissioners who also run polling stations
- ICT kept outside election-operational accounts
- Evidence that can later support disputes — without the system judging them

THE SOLUTION

One platform. Role-based views. Clear technical boundaries.

COMMISSION

Administrative command interface for seven members. Desktop-first. Authentication is not authorization.

ELECTION 1

Eligible students elect delegates from their own electoral unit. School and department are derived, never chosen.

CABINET

Delegates vote a sequential ballot. President + Vice President run as one ticket. One elector, one completed vote.

ICT still exists — for servers, databases, backups, and monitoring — but it is not an election account and cannot enter this portal.

AUTHORITY MODEL

Seven Commission members. One shell. Different actions.

ROLE	PRIMARY RESPONSIBILITY	RESTRICTED POWERS
Chair	Overall Commission authority	Results authorization · emergency initiation
Vice Chair	Senior co-authorization	Emergency second signature · cannot certify results
Secretary General	Records, coordination, registers	No final opening / closing / results
Commissioner · Ruaraka	Station election officer	Town / Kitengela / Western functions stay out of reach
Commissioner · Town	Station election officer	Assignment, not a second account type
Commissioner · Kitengela	Station election officer	Incident log rolls up to the centre
Commissioner · Western	Station election officer	Phone access is evidence upload only

ELECTORAL DESIGN

Two connected elections — not two ballots on one page.

Election 1 · Electoral College

Electorate: eligible KCA students (mock first-year register in this prototype).

They vote only for approved delegate candidates in their own campus / school / department unit.

Delegate seats stay unconfigured until the Commission enters an authoritative number. Nothing is defaulted to 3.

ELECTION 2 · SAKU CABINET

Electorate: the Electoral College Register, generated automatically from confirmed Election 1 winners.

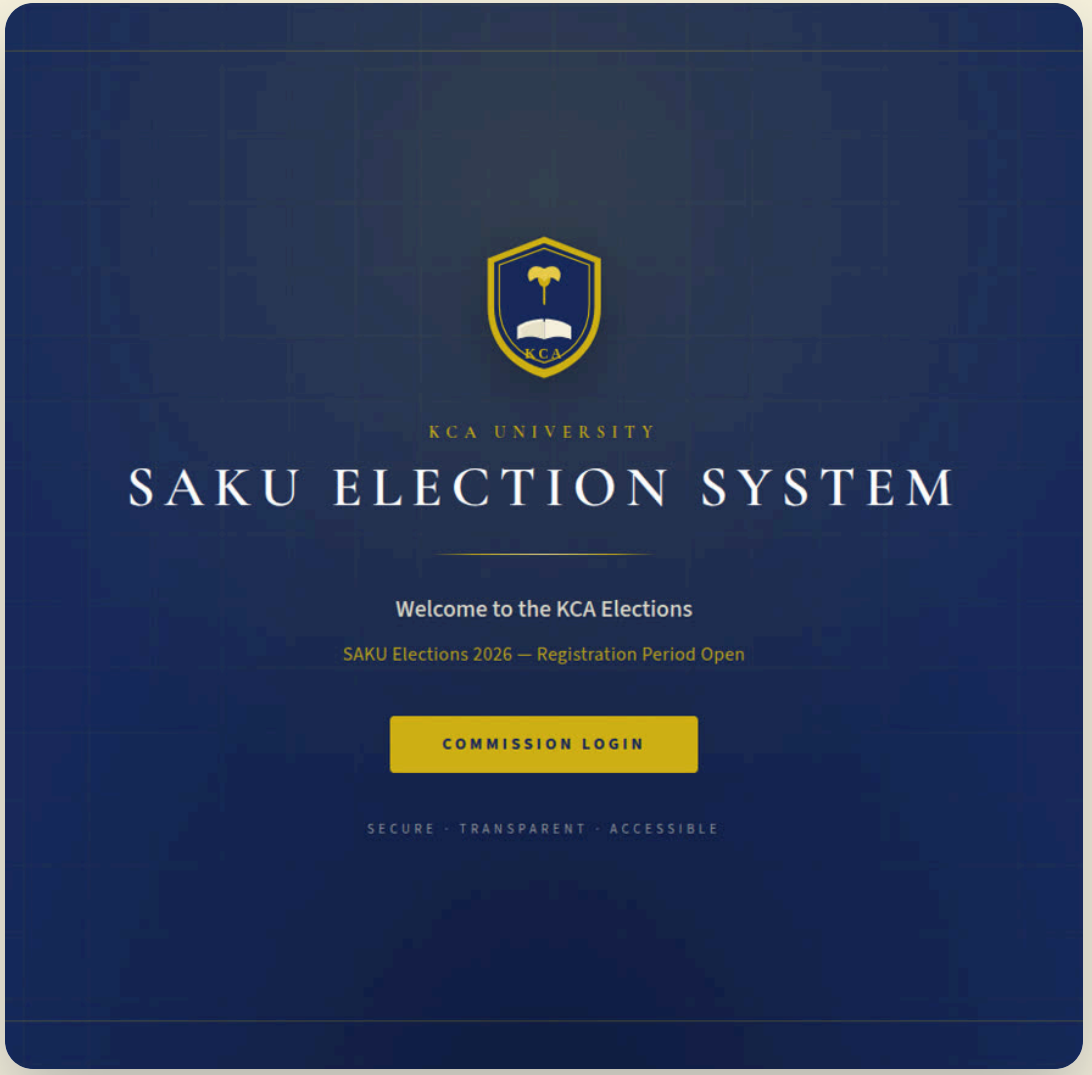
No manual “add elector” workflow. Registration number + OTP. Sequential contests. Joint presidential ticket. Vote locks on final submit.

LANDING

What is this system, and where do I enter?

The public page stays deliberately small. Status is dynamic — registration, voting, or results pending — without becoming a public results dashboard.

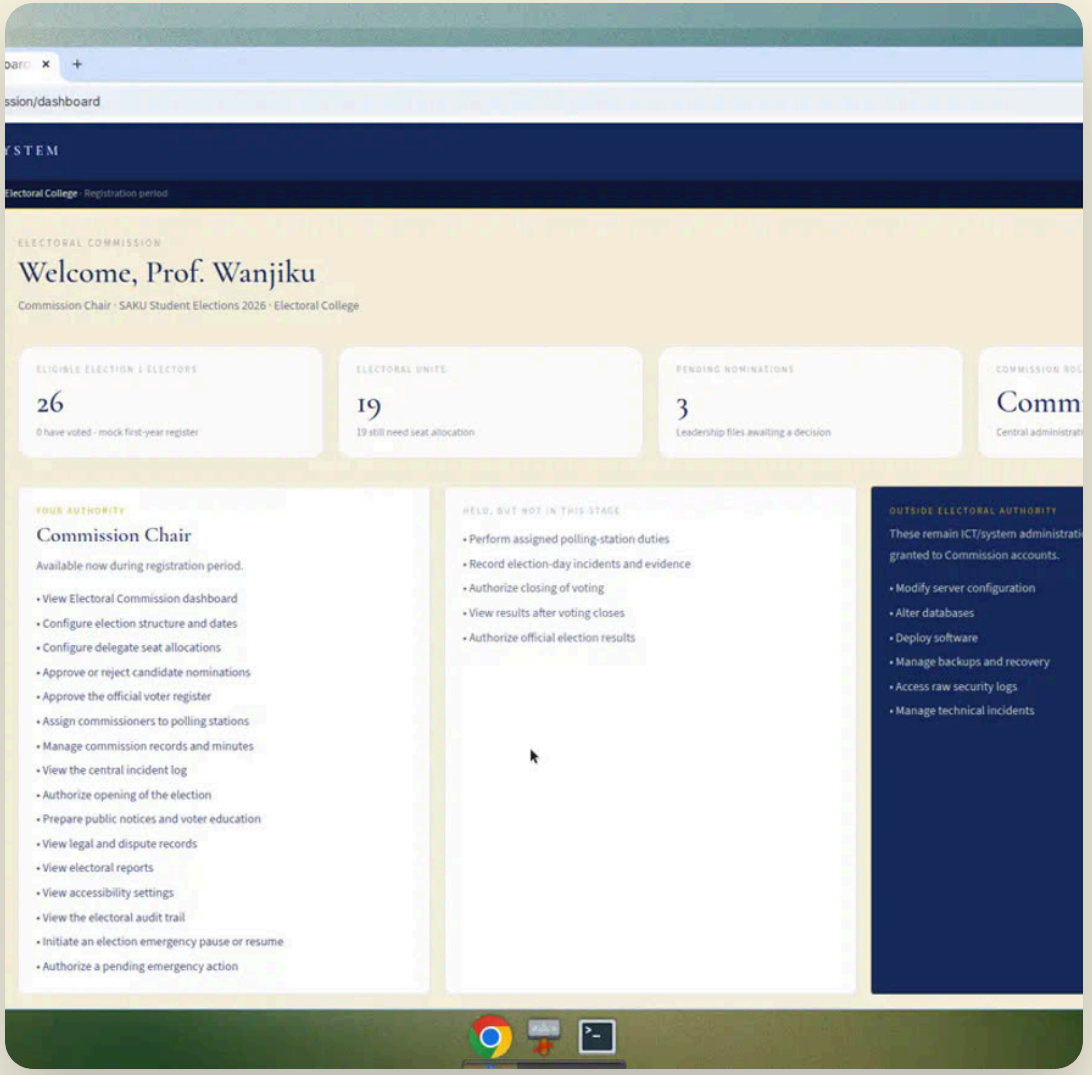
Brand colours are KCA deep blue **#182B5C** and gold **#D0B216**.



COMMISSION PORTAL

A command centre, not a website theme.

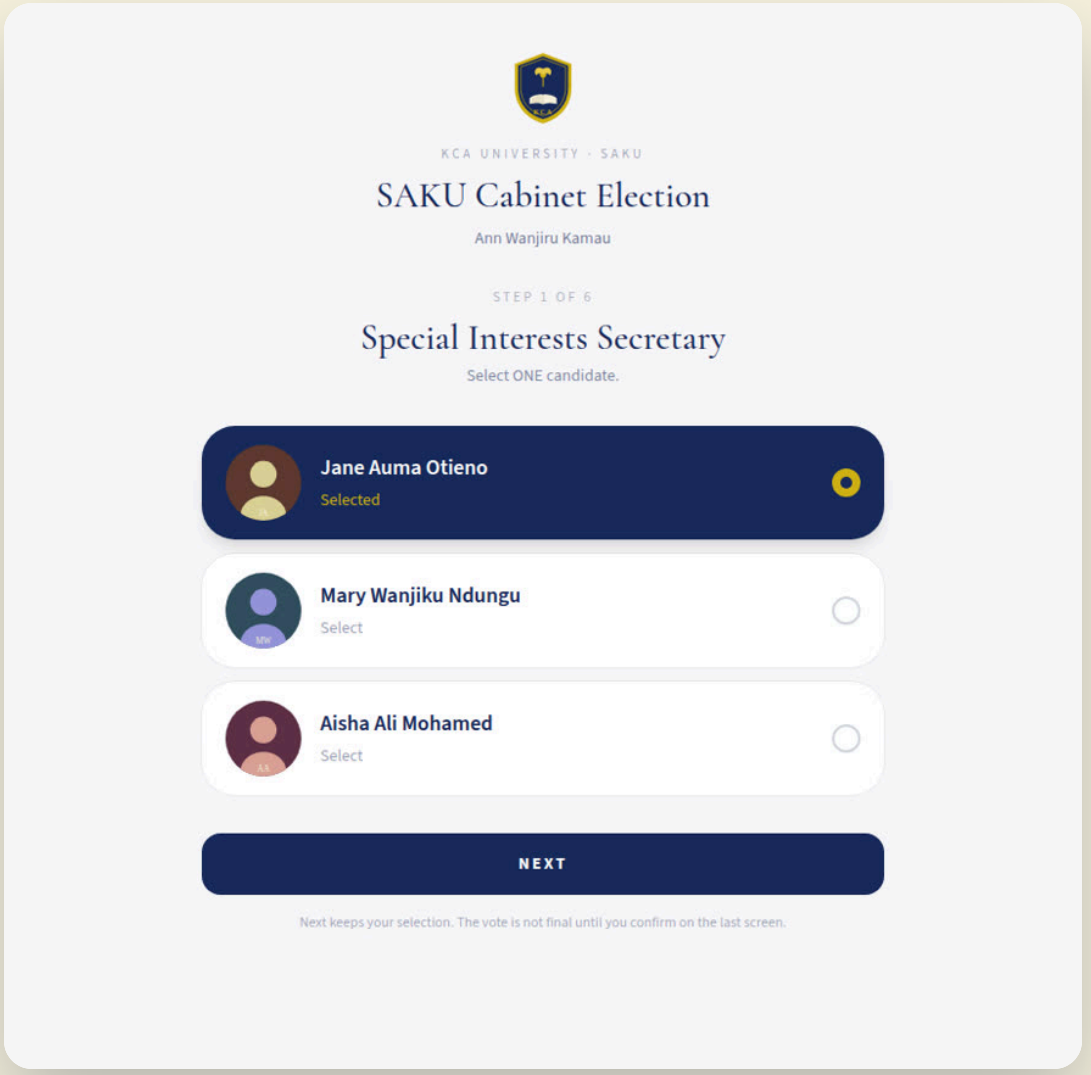
Persistent sidebar. Election and stage in the top bar. Emergency stays a restricted control for Chair and Vice Chair. Results are reached through the election workflow, not as another hamburger item.



CABINET BALLOT

Simple to vote. Impossible to treat as a public poll.

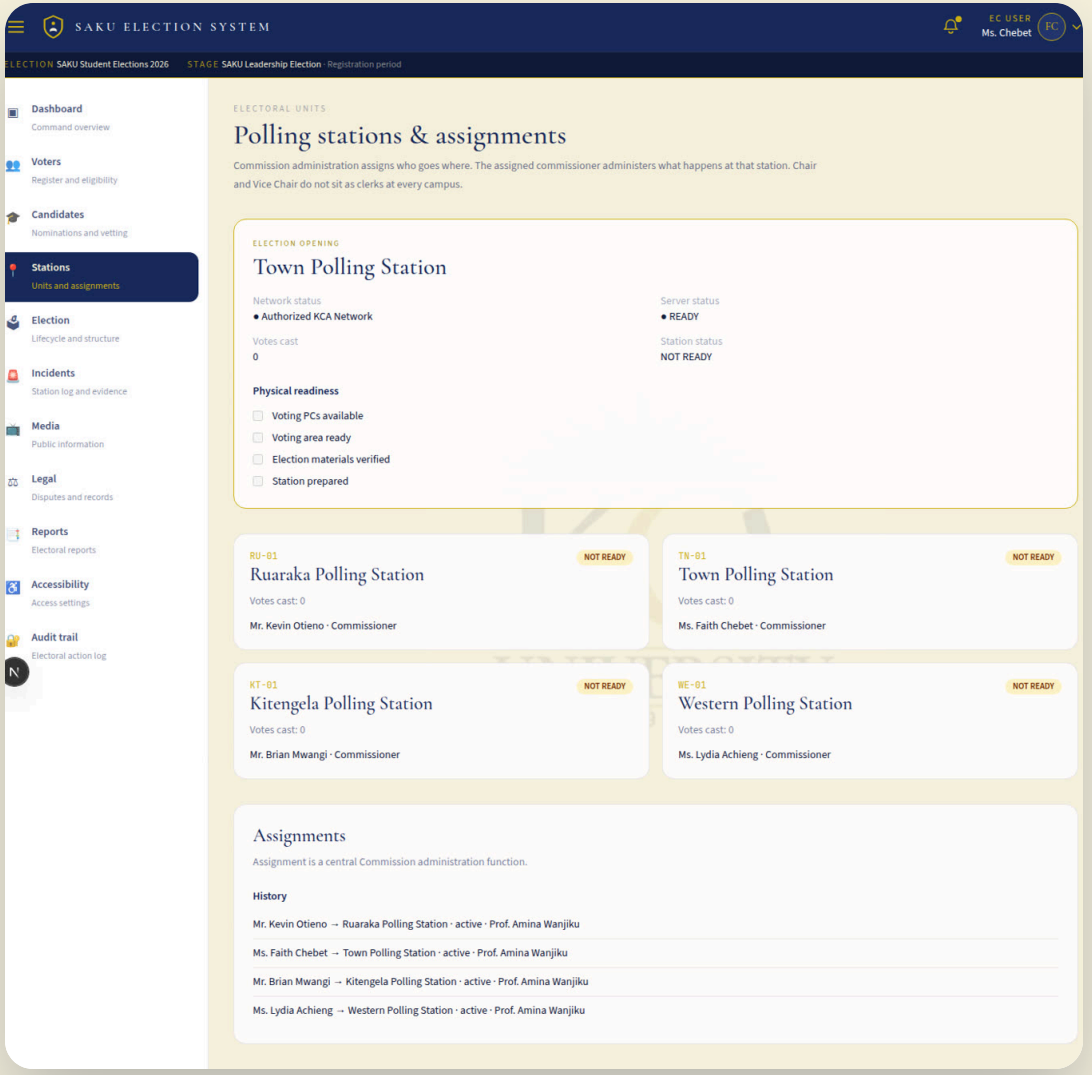
Large selectable cards. Navy selected state. One contest at a time. NEXT keeps a choice; it does not submit. The success screen never shows who was selected.



STATIONS

Commissioners are the clerks.

Assignment activates election-officer functions for Ruaraka, Town, Kitengela, and Western. Physical readiness, then open, monitor, record incidents, close. The system already knows which station the commissioner is reporting from.



TRUST

Security that can be demonstrated in a viva.

Work ID + password + OTP

Institutional credentials, not a hard-coded National ID scheme.

Permission check

Identity → role → election assignment → stage → access.

High-risk re-auth

Opening, closing, results, and emergency require a fresh password.

Two-person emergency

Chair initiates, Vice Chair authorizes. The initiator cannot sign their own pause.

ICT denial

Technical staff may authenticate and still be refused this portal.

Audit trail

Electoral actions, structural changes, and evidence references — not raw server logs.

DEMONSTRATION ACCESS

Try the prototype

Password for every seeded directory account: KCAU@2026

WORK ID	NAME	WHAT YOU WILL SEE
KCAU-EC-001	Prof. Amina Wanjiku	Chair · full electoral authority
KCAU-EC-005	Ms. Faith Chebet	Town station officer · no Emergency item
KCAU-ICT-001	Eng. Peter Kamau	Credentials verify · Commission access denied

Election 1 desk uses Student ID + PIN SAKU2026. Cabinet desk unlocks after the Electoral College Register is generated from Election 1 results.

Secure · Transparent · Accessible

SAKU Election System · KCA University

The system preserves the operational record. It does not adjudicate disputes. It does not invent delegate seats. It does not treat a successful login as permission to run the election.

THANK YOU